

Redpwn 2020 Crypto - 4k-rsa

deuterium — 2020-06-27

Canonical HTML: <https://deut-erium.github.io/WriteUps/2020/redpwn/crypto/4k-rsa/2020-06-27-redpwn-2020-4k-rsa.html>

Abstract

CTF writeup: A large RSA modulus made of smaller primes is factorization's bulk discount in redpwnCTF 2020's 4k-rsa. Recover the factors and decryption exponent.

4k-rsa

Description

Only n00bz use 2048-bit RSA. True gamers use keys that are at least 4k bits long, no matter how many primes it takes...

Files

[4k-rsa-public-key.txt](#) which contains a n , e , c triple

Seems like there are a lot of primes in the factorization of n , since the factorization process is influenced directly by the size of prime factors and not the size of the number being factored itself, it should be fairly doable by [alpetron.ar](#)

[Alpetron](#) > [Programs](#) > Integer factorization calculator

Press the **Help** button to get help about this application. Press it again to return to the factorization. Keyboard users can press CTRL+ENTER to start factorization. This is the WebAssembly version.

• 5 028492 424316 659784 848610 571868 499830 635784 588253 436599 431884 204425 304126 574506 051458 282629 520844 349077 718907 065343 861952 658055 912723 193332 988900 049704 385076 586516 440137 002407 618568 563003 151764 276775 720948 938528 351773 075093 802636 408325 577864 234115 127871 390168 096496 816499 360494 036227 508350 983216 047669 122408 034583 867561 383118 909895 952974 973292 619495 653073 541886 055538 702432 092425 858482 003930 575665 792421 982301 721054 750712 657799 039327 522613 062264 704797 422340 254020 326514 065801 221180 376851 065029 216809 710795 296030 568379 075073 865984 532498 070572 310229 403940 699763 425130 520414 160563 102491 810814 915288 755251 220179 858773 367510 455580 835421 154668 619370 583787 024315 600566 549750 956030 977653 030065 606416 521363 336014 610142 446739 352985 652335 981500 656145 027999 377047 563266 566792 989553 932335 258615 049158 885853 966867 137798 471757 467768 769820 421797 075336 546511 982769 835420 524203 920252 434351 263053 140580 327108 189404 503020 910499 228438 500946 012560 331269 890809 392427 093030 932508 389051 070445 428793 625564 099729 529982 492671 019322 403728 879286 539821 165627 370580 739998 221464 217677 185178 817064 155665 872550 466352 067822 943073 454133 105879 256544 996546 945106 521271 564937 390984 619840 428052 621074 566596 529317 714264 401833 493628 083147 272364 024196 348602 285804 117877 (1225 digits) = 9 353689 450544 968301 × 9 431486 459129 385713 × 9 563871 376496 945939 × 9 734621 099746 950389 × 9 736426 554597 289187 × 10 035211 751896 066517 × 10 040518 276351 167659 × 10 181432 127731 860643 × 10 207091 564737 615283 × 10 435329 529687 076341 × 10 498390 163702 844413 × 10 795203 922067 072869 × 11 172074 163972 443279 × 11 177660 664692 929397 × 11 485099 149552 071347 × 11 616532 426455 948319 × 11 964233 629849 590781 × 11 992188 644420 662609 × 12 084363 952563 914161 × 12 264277 362666 379411 × 12 284357 139600 907033 × 12 726850 839407 946047 × 13 115347 801685 269351 × 13 330028 326583 914849 × 13 447718 068162 387333 × 13 554661 643603 143669 × 13 558122 110214 876367 × 13 579057 804448 354623 × 13 716062 103239 551021 × 13 789440 402687 036193 × 13 856162 412093 479449 × 13 857614 679626 144761 × 14 296909 550165 083981 × 14 302754 311314 161101 × 14 636284 106789 671351 × 14 764546 515788 021591 × 14 893589 315557 698913 × 15 067220 807972 526163 × 15 241351 646164 982941 × 15 407706 505172 751449 × 15 524931 816063 806341 × 15 525253 577632 484267 × 15 549005 882626 828981 × 15 687871 802768 704433 × 15 720375 559558 820789 × 15 734713 257994 215871 × 15 742065 469952 258753 × 15 861836 139507 191959 × 16 136191 597900 016651 × 16 154675 571631 982029 × 16 175693 991682 950929 × 16 418126 406213 832189 × 16 568399 117655 835211 × 16 618761 350345 493811 × 16 663643 217910 267123 × 16 750888 032920 189263 × 16 796967 566363 355967 × 16 842398 522466 619901 × 17 472599 467110 501143 × 17 616950 931512 191043 × 17 825248 785173 311981 × 18 268960 885156 297373 × 18 311624 754015 021467 × 18 415126 952549 973977

Number of divisors: 18 446744 073709 551616

It took about half an hour to factor, one may engage to other activities or alternatively try if the factors are available on [factordb](#).

Anyways, once finished factoring, alpetron produces both the [factors](#) and the Euler's totient ϕ which will be used to compute d

```
d = pow(e,-1,phi) # on python3.8
# or gmpy2.invert(e,phi)
m = pow(c,d,n)
print(bytes.fromhex(hex(m)[2:]).decode())
```

And hurray, we have our flag

flag-t0000'm4nyyyy'pr1m355555"

| Interactive controls are omitted from this PDF. See the canonical HTML version.